CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 5: Arithmetic Progressions

Previous year board questions, topic-wise: 24 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Common difference and the nth term

1. Find the common difference of the AP $\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, 4\sqrt{2}, \ldots$ (a) $\sqrt{2}$ (b) 1 (c) $2\sqrt{2}$ (d) $-\sqrt{2}$ CBSE 2026, 1 mark

2. The numbers $p-1$, $p+1$ and $2p+3$ are in AP. Find p. (a) -2 (b) 4 (c) 0 (d) 2 CBSE 2023, 1 mark

3. How many two-digit numbers are divisible by 3? CBSE 2019, 1 mark

4. Which term of the AP $3, 15, 27, 39, \ldots$ is 120 more than its 21st term? CBSE 2019, 2 marks

5. In an AP the common difference is -4 and the seventh term is 4. Find the first term. CBSE 2018, 1 mark

6. Four consecutive terms of an AP add up to 32. The product of the first and last of them is to the product of the two middle ones as $7 : 15$. Find the four numbers. CBSE 2018, 4 marks

7. For what value of k are $k+9$, $2k-1$ and $2k+7$ consecutive terms of an AP? CBSE 2016, 1 mark

8. The 4th term of an AP is zero. Prove that its 25th term is three times its 11th term. CBSE 2016, 2 marks

2 Case study

9. In a potato race, 10 potatoes lie in a straight line. The first is 5 m from a bucket at the starting point and the rest are 3 m apart. A runner starts at the bucket, fetches the nearest potato, runs back to drop it in the bucket, and repeats this for each potato in turn. (i) How far does the runner go to collect the first potato? (ii) How far to collect the second? (iii) Find the total distance run, OR the time taken for the whole race at an average speed of 5 m/s. CBSE 2026, 4 marks

10. In a fund-raising run, the first round is 300 m long and every round after it is 50 m longer than the one before, so the rounds are 300 m, 350 m, 400 m and so on. There are 10 rounds in all. (i) Write the 4th, 5th and 6th terms of this AP. (ii) How long is the 8th round? (iii) Find the total distance covered in all 10 rounds, OR the total distance covered by a runner who stops after 6 rounds. CBSE 2025, 4 marks

11. A matchstick pattern is built in stages. Figure 1 is a square with a triangle on each of its four sides. Each new figure adds one more square to the row, with triangles on its free sides. Figures 1, 2 and 3 contain 4, 6 and 8 triangles, and use 12, 19 and 26 matchsticks. (i) Write the AP for the number of triangles and its nth term. (ii) Which figure uses 61 matchsticks? CBSE 2022 Term 2, 4 marks

3 Sum of the first n terms

12. The sum of the first m terms of an AP is $2m^2 + 3m$. Find its second term. CBSE 2025, 1 mark

13. An A.P. has first term $a = 7$ and nth term $a_n = 84$, and its first n terms add up to $\frac{2093}{2}$. Find n. (a) 22 (b) 24 (c) 23 (d) 26 CBSE 2024, 1 mark

14. In an A.P., the first and eighth terms have sum 32 and product 60. Find the first term and the common difference, and then the sum of the first 20 terms. CBSE 2024, 5 marks

15. An A.P. has 40 terms. Its first 9 terms add up to 153 and its last 6 terms add up to 687. Find the first term, the common difference and the sum of all 40 terms. CBSE 2024, 5 marks

16. In an AP, the 11th term and the 17th term are in the ratio $3 : 4$. Find the ratio of the 5th term to the 21st term, and the ratio of the sum of the first 5 terms to the sum of the first 21 terms. CBSE 2023, 5 marks

17. 250 logs are stacked in rows: 22 in the bottom row, 21 in the next, 20 in the one above, and so on. How many rows are there, and how many logs are in the top row? CBSE 2023, 5 marks

18. Find the sum of the first 30 terms of the AP $-30, -24, -18, \ldots$ CBSE 2022 Term 2, 2 marks

19. The sum of the first n terms of an AP is $S_n = n(4n + 1)$. Find the AP. CBSE 2022 Term 2, 2 marks

20. The sum of the first n terms of an AP is $S_n = 3n^2 - 4n$. Find its nth term. CBSE 2019, 2 marks

21. The first four terms of an AP add up to 40 and the first 14 terms add up to 280. Find the sum of its first n terms. CBSE 2019, 4 marks

22. Find the sum of the first 8 multiples of 3. CBSE 2018, 2 marks

23. The sums of the first n terms of two APs are in the ratio $(7n + 1) : (4n + 27)$. Find the ratio of their mth terms. CBSE 2016, 3 marks

24. The houses in a row are numbered from 1 to 49. Show that there is a house number X for which the sum of the numbers before it equals the sum of the numbers after it. CBSE 2016, 4 marks

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